

MATH230: Tutorial Two

Propositional Logic: Natural Deductions with Negation

Key ideas

- Write natural deduction proofs involving $\neg$ negation.
- Write natural deductions using Ex Falso i.e. $\perp$ elimination.

Relevant Topic: Propositional Logic

Relevant reading: L $\exists$ VN Chapters 3,4 and Simon section 1.1

Hand in exercises: 1a, 1b, 1c, 1i, 1k

Due Friday @ 5pm to the submission box on Learn.

Discussion Questions

1. Show $A \vdash \neg\neg A$.

2. Show $(A \vee B) \wedge (A \vee C) \vdash A \vee (B \wedge C)$.

Tutorial Exercises

1. **Minimal Logic.** Provide natural deduction proofs of the following sequents. These deductions require only the use of minimal logic; the introduction and elimination rules for $\wedge$ conjunction, $\vee$ disjunction, $\rightarrow$ implication, and the definition of $\neg$ negation as an implication.

- (a) $\neg A \vdash (C \rightarrow A) \rightarrow \neg C$ [Handin exercise]
- (b) $A \wedge \neg B \vdash \neg(A \rightarrow B)$ [Handin exercise]
- (c) $A \rightarrow C, B \rightarrow D, \neg C \vee \neg D \vdash \neg A \vee \neg B$ [Handin exercise]
- (d) $A, \neg A \vdash \neg B$
- (e) $A \rightarrow B, A \rightarrow \neg B \vdash \neg A$
- (f) $A \rightarrow \neg B \vdash B \rightarrow \neg A$
- (g) $\neg(A \wedge B) \vdash A \rightarrow \neg B$
- (h) $A \vdash \neg\neg A$
- (i) $\neg\neg\neg A \vdash \neg A$ [Handin exercise]
- (j) $\neg A \vee \neg B \vdash \neg(A \wedge B)$
- (k) $\neg A \wedge \neg B \vdash \neg(A \vee B)$ [Handin exercise]
- (l) $\neg(A \vee B) \vdash \neg A \wedge \neg B$
- (m) $A \rightarrow \neg B \vdash \neg(A \wedge B)$
- (n) $\vdash \neg\neg(A \vee \neg A)$ [Challenge!]

2. **Intuitionistic derivations.** Provide natural deduction proofs of the following. You do not need to use the *classical* $\perp$ rule for these questions, but may find that the *intuitionistic* $\perp$ rule is necessary.

- (a) $A, \neg A \vdash B$
- (b) $\neg A \vdash A \rightarrow B$
- (c) $\neg A \vee B \vdash A \rightarrow B$
- (d) $A \vee B, \neg A \vdash B$
- (e) $\vdash \neg(B \rightarrow A) \rightarrow (A \rightarrow B)$
- (f) $A \rightarrow B, A \rightarrow \neg B \vdash A \rightarrow C$
- (g) $A \vee B, \neg A, \neg B \vdash C$